

Shandong Chenxuan Robot Technology Co., Ltd. User Manual

Shandong Chenxuan Robot Technology Co., Ltd. User Manual

ShandongChenxuanRobotTechnologyCo.,Ltd.U serManual



Manual version: V1.1.0

Applicable Product Model: DYZJ_WCR_01

Applicable Software Version: V1.2.3.RedFire

Shandong Chenxuan Robot Technology Co., Ltd. User Manual

1. Safety	1
2. Transportation and Hoisting	1
3. Hardware Components and Installation	1
3.1 Overview by Figures	1
3.2 Hardware Clusters:	3
3.3. Installation	3
4. Basic Software Overview	3
4.1 Program Interface	3
4.2 Craft Interface	5
4.3 Setting Interface	6
4. 4 SmartPlus Interface	7
5. Power-On Check and Preparation	8
5.1 Power-On Checks	8
5.2 Pre-Work Preparation	8
6. Operating Guide	9
6.1 Preparations	9
6.2 Teaching	9
6.3 Welding Modes	10
6.4 Welding craft	12
7. Ending work and storage	14
7.1 Ending Work	14
7.2 Storage Conditions	14
8. Vulnerable Parts and Maintenance	14
8.1 List of Vulnerable Parts	14
8.2 Maintenance and inspection plan	14
9. Fault list and Solutions	15
9.1 List of faults and solutions other than welding machines	15
9.2 MEGMEET Ehave2 CM500R Welding machine failure list and solutions	16
10. SmartPlus Components	20
10.1 Intelligent Dragger	20

Revisions

[illegible]

1. Safety

1. During transportation, ensure quality and center of gravity. No personnel should be under equipment during hoisting.
2. Beware of mobile cart's wheel hazards during transportation.
3. Prevent robot or wire feeder from tipping over during movement.
4. Only qualified personnel should connect the power.
5. The equipment must be properly connected in accordance with the AC 380V three-phase five-wire system. Especially the ground wire, which must be reliably connected and is strictly prohibited from being left unconnected.
6. Do not open the control box while powered on due to high voltage.
7. Welding operators must be professionally trained and certified by Dongyun Robotic.
8. Maintenance personnel must be professionally qualified and certified by Dongyun Robotic.
9. Do not disassemble or modify the device without permission. For integration with other systems (e.g., MES), consult Dongyun Robotic.
10. Any safety information contained in this manual should not be considered a general safety guarantee. Even if all safety instructions are followed, personal injury or equipment damage may still occur.

2. Transportation and Hoisting

1. Long-distance transportation must be packed in wooden boxes to prevent tipping and bumping.
2. Short-distance internal movement requires a forklift and pallet. To prevent tipping and bumping. Robot and wire feeder tipping risks must be minimized. Forklift speed limit: 5 km/h.
3. Lifting is limited to welding machines and wire feeders. Lifting of robots and other devices is strictly prohibited. There are two special lifting rings on the top of the welding machine. Be sure to use them for lifting.

3. Hardware Components and Installation

3.1 Overview by Figures

The complete system is divided into three parts:

Part 1: Welder and cart near the power connection.

Part 2: Control box and wire feeder aligned with the robot.

Part 3: Robot and welding torch at the weld location.



Figure 3. 1 Hardware components

Components Included:

1. Power connector (380V, 3-phase, 5-wire) with circuit breaker inside.
2. Wire feeder.
3. Welding machine with a individual breaker
4. Robot.

5. SmartPlus component, Intelligent dragger.
6. Welding torch.
7. Quick-change magnetic base: first press down the handle, then rotate the handle clockwise or counterclockwise to absorb or release. The specific direction refers to the red surface mark of the magnet.
8. Control cabinet, includes antenna connector. When in use, connect the antenna to the SMA connector on the upper right corner of the control cabinet.
9. Mobile cart
10. Bracket
11. Tablet/PDA, with pre-installed "Dongyun Welding Control" APP, which is to control the robot to do welding.

3.2 Hardware Clusters:

1. Robot cluster: robot, quick-change magnetic base, welding torch assembly.
2. Welding machine cluster: welding machine, wire feeder, water tank, wire feeding tube, welding connection bracket wire feeder connection bracket, integrated line pipe, mobile cart, mobile cart-bracket.
3. Control cabinet cluster: control cabinet, power connector, main power switch, robot cable, power cord, antenna.

3.3. Installation

1. Device is pre-installed and verified. User only needs to connect the power (AC 380V, 3-phase 5-wire) and gas pipe. And it is necessary to check whether the cable voltage is correct and whether the gas pipe is correctly connected to the inert gas. The power connection requirements: blue N pole (neutral line), yellow-green PE (ground), and the other three colors are connected to L1, L2, and L3 respectively (no phase sequence requirements).
2. Take the robot out of the foam box and use the magnetic base to attach the robot to the flat surface where the robot is temporarily placed. All three magnets on the magnetic base must be opened without missing any.
3. Move the robot and wire feeder to the work site after Chapter 5 checks.

4. Basic Software Overview

4.1 Program Interface

After opening the "Dongyun Welding Control" APP, and the Program page shows as Figure 4.1:

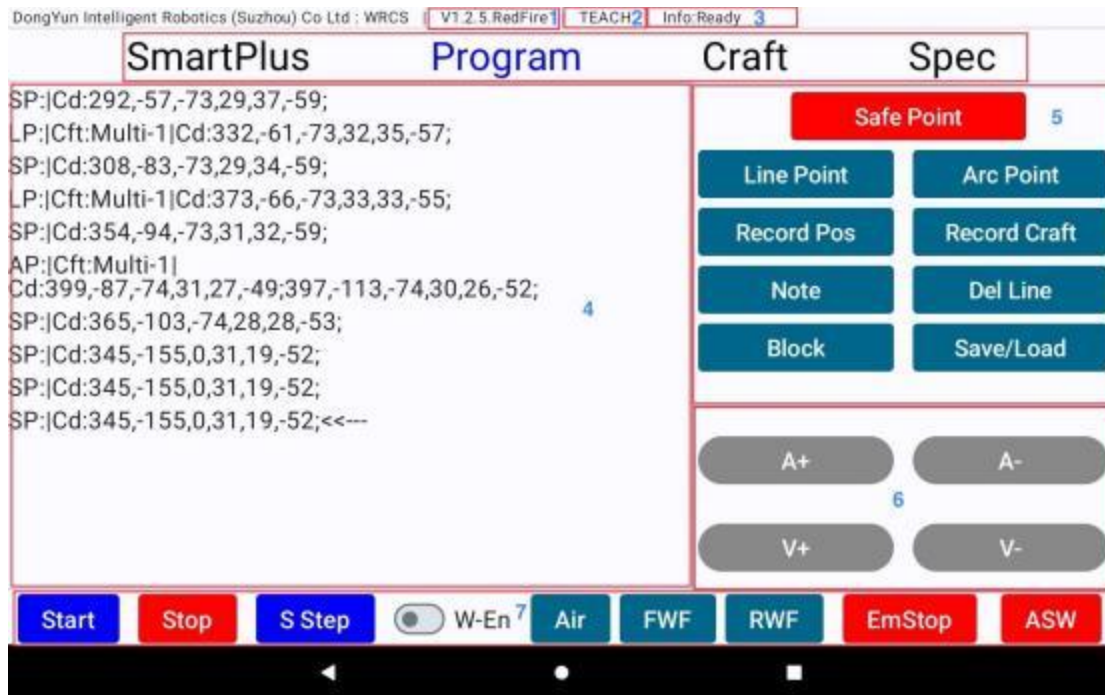


Figure 4.1 Program interface function description

Area 1: Shows software version

Area 2: status (teaching or running)

Area 3: Errors, (for mor details, refer to Chapter 9)

Area 4: Program steps, you can select the corresponding line by clicking on it.

Area 5: Programming Button:

"Safe point" – to insert non-welding position points.

"Line Point" – to insert straight line welding position points.

"Arc Point" – to insert arc welding position points.

"Record Pos" – to record the position information of the above three types of points.

"Record Craft" – to record the welding craft packages of "Line Point" and "Arc Point" which are the two points that need to be welded.

"Note" – to insert a line of text information to facilitate users to understand the programming progress.

"Del Line" – to delete the selected line of program.

"Block" – to insert a line of stay time (in milliseconds) instructions in place.

"Save/Load" – to save the current program to any one of the three storage spaces or extract and overwrite the current program from any one of the three storage spaces.

Area 6: Used to adjust current and voltage in real time during welding.

Area 7: Function buttons:

"Start" -- start executing the program.

"Stop" -- stop the program and try to clear existing errors.

"S Step" -- execute the selected line of the program in single step.

"W-En" -- switch between the three states of fast simulation/full function simulation/welding.

"Air" -- check the gas.

"FWF" -- feed the wire.

"RWF" -- retract the wire.

"EmStop" -- immediately terminate all actions.

"ASW" -- directly start the arc to assist in the temporary assembly of workpieces.

4.2 Craft Interface

After opening "Dongyun Welding Control" APP, click the "Craft" tab to open the craft interface. The craft package needs to be connected to the device to read it. The interface is shown in Figure 4.2 below:



Figure4.2 Craft interface function description

Area 1: Tap the downward arrow on the right to view all the craft packages in the current device. Select the craft package you want to view in the pop-up list and tap it to display the content of the craft package.

Area 2:

"New" -- to create a custom-named craft package in the internal storage

space of the device.

“Save” – to overwrite the original parameters with the modified parameters.

“Save as” – to copy the craft to a new craft package.

“Del” – to delete the specified craft package from the internal storage space of the device.

Area 3: Used to adjust the parameters of the three-stage welding craft: arc starting, normal welding, and arc termination. Slide your finger up and down in this area to view the complete parameter list.

Area 4: Used to adjust the arc swing parameters. More parameters can be seen after turning on the arc swing function.

Area 5: It is used to adjust the multi-line craft parameters. More parameters can be seen after turning on the multi-line function.

Area 6: Used to adjust intermittent welding craft parameters. More parameters can be seen after turning on the intermittent welding function.

4.3 Setting Interface

After opening the "Dongyun Welding Control" APP, click the "Settings" tab to open the settings interface. The content modified in the settings interface needs to be connected to the device to be effective. The interface is shown in Figure 4.3 below:



Figure 4.3 Setting interface function description

The area where the blue number 1 is located: Set the current installation direction of the robot. Enter 0 for upright installation, 1 for side mounting, and 2 for upside

down mounting.

The area where the blue number 2 is located: Set the real-time current adjustment button described in the "Program" interface during welding to adjust the amount of adjustment each time it is clicked.

The area where the blue number 3 is located: Set the real-time voltage adjustment button described in the "Program" interface during welding to adjust the adjustment each time.

The area where the blue number 4 is located: Set whether to automatically clear the current program after welding is completed. Fill in 0 to clear the program after welding is completed and fill in 1 to not clear the program after welding is completed.

4. 4 SmartPlus Interface

After opening the "Dongyun Welding Control" APP, click the "SmartPlus" label to open the SmartPlus component setting interface. The content modified in the setting interface needs to be connected to the device to be effective. The interface is shown in Figure 4.4 below:

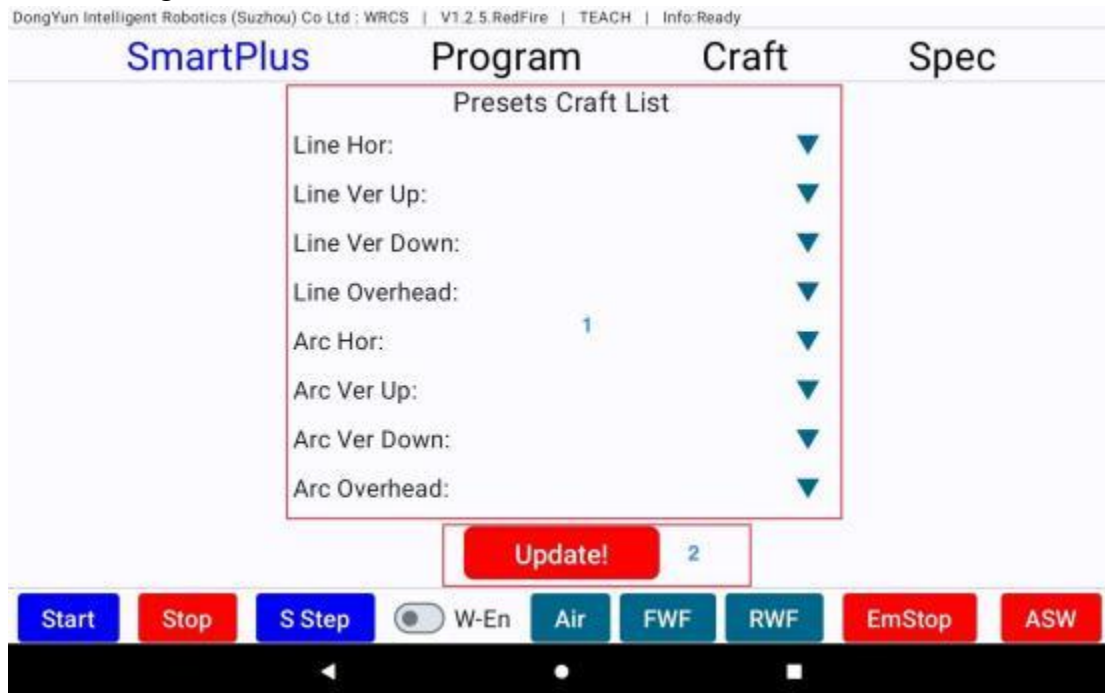


Figure 4.4 SmartPlus component interface function description

Area 1: Set which craft package to use when the intelligent dragger recognizes that the craft needs to be used through spatial position.

Area 2: Update the above craft package mapping list to the device.

5. Power-On Check and Preparation

5.1 Power-On Checks

1. Turn on the main power switch. Within 10 seconds after power-on, the fan of the control cabinet should rotate at the highest speed for some time to determine whether the exhaust and supply fans are working normally.

2. 10 seconds after power-on, the AC contactor in the control cabinet will be energized, the robot will be powered on, and within the next 45 seconds, the indicator light at the end of the robot's 6th axis will flash until the green light is on steadily, to determine whether the robot is working normally.

3. Turn on the tablet/PDA, open the "Dongyun Welding Control" APP, and the prompt should display "Ready" to determine that the system is working normally. Otherwise, please refer to Chapter 8 of this manual for solutions.

4. The device must be placed in an operating environment below 45 degrees Celsius. If it exceeds 45 degrees Celsius, the robot will never power on.

5. Check whether the water level in the welding machine's water tank is at the MAX position and whether it needs to be replaced with special antifreeze in winter.

6. Check whether the water circulation of the welding machine is normal and whether the welding machine alarms. If the welding workpiece plane and the welding machine plane are not on the same plane, water circulation failure is more likely to occur.

5.2 Pre-Work Preparation

1. The integrated wire tube must be straightened, not stacked into a coil, and not hung on the horn hanging.

2. Check the remaining amount of welding wire and change the wire in advance. If there is no welding wire during welding, there is a high probability that it will melt and block the contact tip, burn out the water circulation system and cause water leakage, etc.

3. Do not forget to connect the welding circuit correctly and reliably.

4. Prepare the craft. Call the general craft package provided by Dongyun Robotics and adjust the craft appropriately according to the actual situation by conducting actual tests on the test board.

6. Operating Guide

6.1 Preparations

1. Check the gas. Press the "Air" button on the software interface. After the electronic valve opens, the tip of the welding gun should clearly feel the airflow and sound. The air in the hose must be completely exhausted, usually for about 2 minutes.

2. Feed wire. Press the "FWF" button on the software interface, and the wire feeder rotates clockwise to push out the wire. If there is no wire in the wire tube, the wire feeding time will be longer. Finally, the electrode extension is guaranteed to be 8mm. Be sure to keep this consistent when dragging backwards for teaching.

3. Wire retraction: If the welding wire extends too far from the tip of the gun, you can press the "RWF" button on the software interface, and the wire feeder will rotate counterclockwise to retract the welding wire. Generally, you can directly use diagonal pliers to cut the welding wire. It is not easy to control the electrode extension using the wire retraction mechanism.

4. Applying a special welding torch cleaning agent to the welding torch is beneficial to extending the life of the contact tip and the nozzle and is beneficial to the formation of the weld.

5. Clean the weld. Grind, preheat, clean and other pretreatment work should be done on the welds to be welded in advance to ensure the welds are formed consistently and the welding craft can be effectively reproduced.

6.2 Teaching

1. Edit a straight weld

I. This step is optional. Drag the robot to 45 degrees and 10 cm above the arc starting position, and press "Safe point" on the software interface.

II. Drag the robot to the arc starting position in the correct welding posture (check the stick out length!), and press "Safe point" on the software interface.

III. Drag the robot to the arc termination position in the correct welding posture (check the stick out length!), press "Straight Point" on the software interface, select the craft package and press "Confirm".

IV. This step is optional. Drag the robot to 45 degrees and 10 cm above the arc termination position, and press "Safe point" on the software interface.

2. Editing an Arc Weld

I. This step is optional. Drag the robot to 45 degrees and 10 cm above the arc starting position, and press "Safe point" on the software interface.

II. Drag the robot to the arc starting position in the correct welding posture (check the stick out length!), and press "Safe point" on the software interface.

III. Drag the robot to the middle point of the arc in the correct welding posture (please make sure to ensure the stick out length!), press "Arc Point" on the software interface, select the craft package and press "Confirm".

IV. Drag the robot to the arc termination position in the correct welding posture (check the stick out length!), and press "Record Pos" -> "P2" on the software interface.

V. This step is optional. Drag the robot to 45 degrees and 10 cm above the arc termination position, and press "Safe point" on the software interface.

3. Edit a combined weld of the above weld

As with the above method, only edit one weld directly at the end of another weld, without any safe points in between.

6.3 Welding Modes

1. Normal welding

During the welding craft, the welding current and voltage can be adjusted in real time on the software interface through the four buttons "A+", "A-", "V+", and "V-". The adjustment amplitude each time is adjusted in the "Spec" page on the software interface.

- I. Drag the robot to near the starting position.
- II. Turn on the "W-En" switch on the software interface.
- III. On the software interface, press "Start".

2. Fast simulation welding

- I. Drag the robot to the starting position.
- II. Turn off the "W-En" switch on the software interface.
- III. Press "Start" on the software interface.

3. Full-featured welding simulation

Compared with fast simulation, full-function simulation welding means that the movements and speeds are the same as those during welding, and the movements also include the torch weaving.

- I. Drag the robot to near the starting position.
- II. Press the “W-En” button once on the software interface. When the button background turns green, it is in full-function simulation state.
- III. Press "Start" on the software interface.

4. Single-step simulation movement

- I. Select the required single-step execution instruction on the software interface.
- II. Press "S Step" on the software interface.

5. Manual spot welding

- I. Align the robot to the arc starting position and release the drag button.
- II. Turn on the “W-En” switch on the software interface.
- III. Press the "ASW" button on the software interface;
- IV. Release the "ASW" button after spot welding is moderate.

6. Special Circumstances

- I. If you need to stop immediately during welding or any movement, you need to press the "Stop" button on the software interface.
- II. If an emergency stop is required, press the "EmStop" button on the software interface, or press the emergency stop button on the side wall of the control cabinet, or press the emergency stop button on the movable emergency stop box.
- III. If the arc fails, the robot will restart the arc three times.
- IV. If you press the "EmStop" button on the software interface, you can press the "EmStop" button again to release the emergency stop and automatically restart the robot. Wait for the status signal to change to "Ready" to start working again.
- V. Before and after each welding, be sure to clean the welding torch head and other parts to improve welding quality and extend service life.
- VI. Before each welding, be sure to check the remaining amount of welding wire in the wire feeder to avoid wire breakage and burning during welding, which may cause serious damage to the welding torch parts.
- VII 、 If you need to pause welding, you need to press the "Pause" button on the software interface; conversely, if you need to resume welding, you need to press the "Continue" button on the software interface.

- VIII、 If you need to save a program, you need to press the "Save/Load" button on the software interface, and then select a location from 1-3 in the pop-up interface to

store it; conversely, if you need to extract a saved program, you need to select one from 1-3 in the pop-up interface to extract it. For example, if you saved the program in location 3 last time, then if you want to restore the last program, just extract it from location 3.

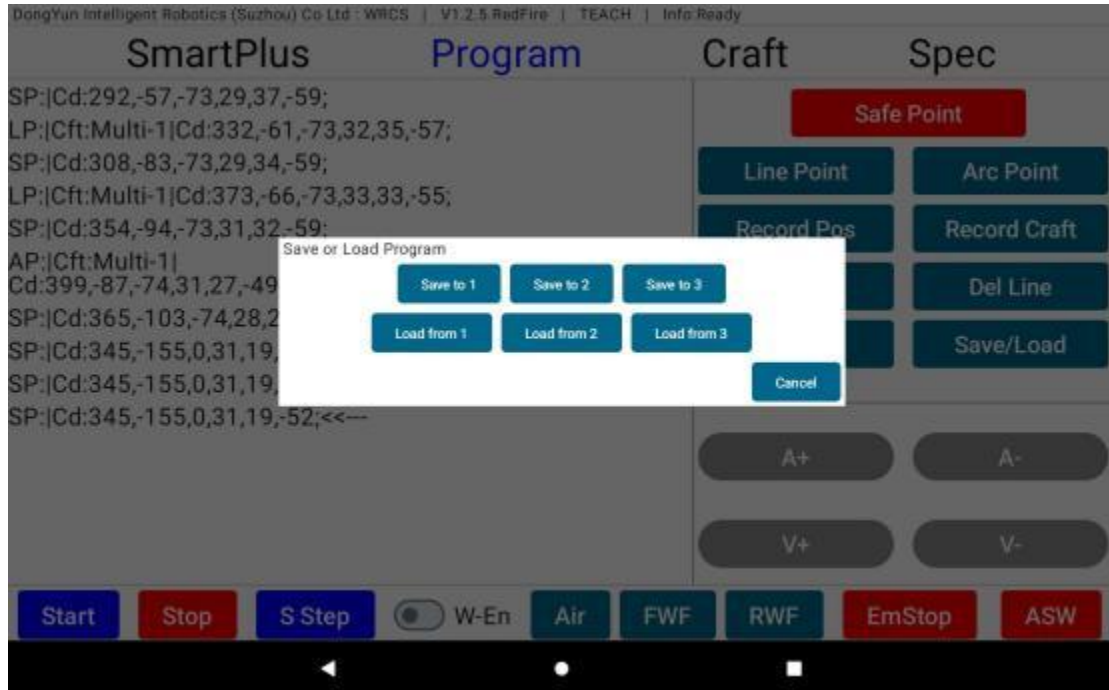


Figure 5.1 Extract or save program description

6.4 Welding craft

1. Three-stage Welding Craft

Under the "Craft" column in the software interface, the craft parameters of each stage of the three-stage welding craft, arc starting → normal welding → arc termination, can be adjusted.

2. Arc Weaving

In the "Craft" column of the software interface, after turning on the arc weaving function, you can adjust the type and various parameters of the arc weaving.

3. Multi-Line Welding

Under the "Craft" column of the software interface, after turning on the multi-pass function, you can adjust the number of channels and the offset of each channel. When using multi-pass welding, please pay attention to creating a new craft package with the corresponding number of channels. The package name ends with "-n", "-" is a minus sign, where $n \geq 1$, n must be continuous, and when calling, be sure

to call the craft package ending with "-1". The multi-layer multi-channel function has a fixed format, please be sure to pay attention to the format as follows:

The first instruction, fixed, must be the arc starting position, safe point.

The second instruction, fixed, must be a straight line or an arc;

The third instruction, fixed, must be the pointing point of the second instruction (replaced by a safe point) to indicate the plane and offset direction.

For the fourth to nth instructions, if there are multiple straight lines or arcs or straight line and arc splicing, please repeat the second and third instructions to add them.

Starting from the n+1 instruction, a return path must be provided, all of which are safe points.

Note that it will stay at the last safe point during operation, please manually clean the slag and maintain the welding torch before pressing the "Continue" button to carry out the next welding.

Here is the example:

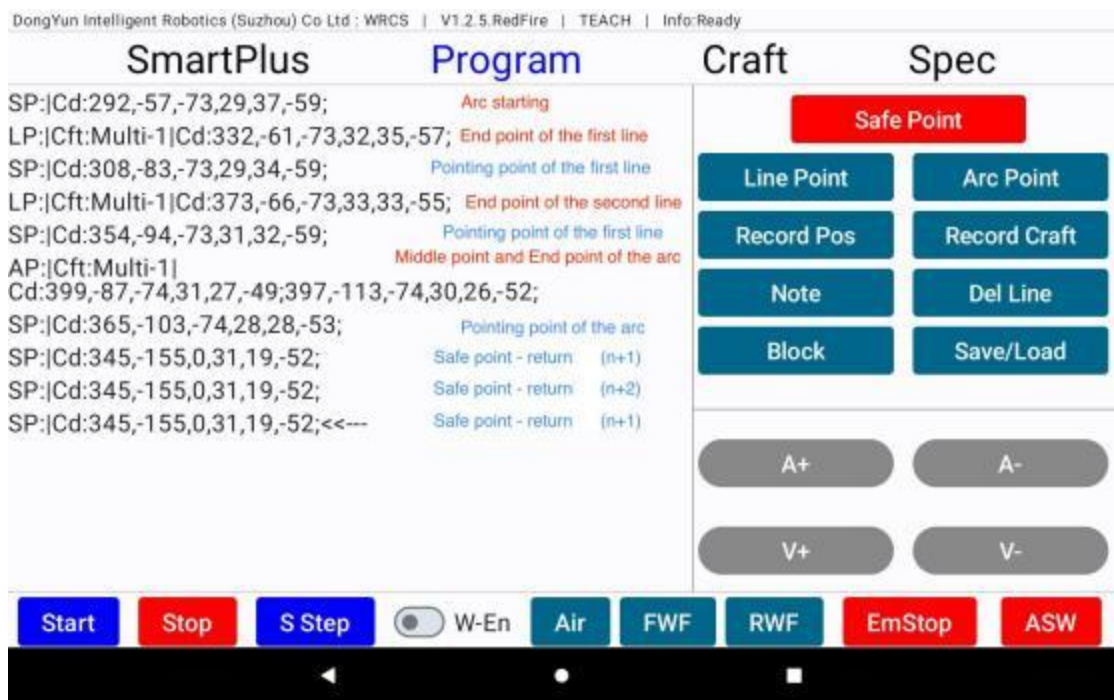


Figure 5.2 Illustration of a concrete example of multi-line welding

4. Intermittent welding

In the "Craft" column of the software interface, after open the "IMT Enable" button, you can adjust the welding length and discontinuity length, and you can adjust whether to treat multiple teaching trajectories as one arrangement, whether the two ends are evenly distributed, and whether to weld if the length to be welded at the

end is less than the set welding length in the case of non-uniform distribution at both ends.

7. Ending work and storage

7.1 Ending Work

1. Drag the robot to an easy-to-store position.
2. Power-off of the whole machine.
3. Storage cables.
4. Place the robot in the special position of the mobile cart.
5. Drag the whole machine to the established position.

7.2 Storage Conditions

1. Must be stored in a location free of rainwater splashing.
2. Must be stored in a non-corrosive location.
3. Must be stored in a dry location.
4. The power must be cut off.
5. The antenna of the control cabinet needs to be unscrewed and placed in the control cabinet.
6. The tablet needs to be charged in time, if it is not used for a long time, it also needs to be taken out and recharged once every 1 month.
7. Ensure that the storage temperature is not lower than -10 degrees Celsius and not higher than 45 degrees Celsius.

8. Vulnerable Parts and Maintenance

8.1 List of Vulnerable Parts

1. Nozzle TRM 550WTE51-45L234H073 1-2A (freely selectable), the default is TRM 015-62-14C.
2. Contact tip TRM 550WTE51-45L234H073 1-3A (freely selectable), the default is TRM 035-12-12A.

8.2 Maintenance and inspection plan

1. Regularly check whether the water volume and water circulation of the water tank of the welding machine are normal, and special antifreeze must be filled in winter.
2. Regularly check whether the fan of the control cabinet is running normally,

especially in summer.

3. Regularly inspect the control cabinet and check whether the internal connection line is loose (pay attention to the danger of high voltage).

4. Regularly check whether the wire feeding tube is damaged and whether the wire feeding is smooth.

5. Regularly replace the dustproof filter of the fan of the control cabinet.

6. Replace the welding torch nozzle kit regularly.

7. Replace the air supply tube regularly.

9. Fault list and Solutions

9.1 List of faults and solutions other than welding machines

DYZJ_WCR_01List of system failures and solutions		
number	Faults	solution
1	Connecting to robot	1 、 Check whether the control cabinet is turned on (whether it is powered on) 2. Check whether the glue stick antenna is connected 3. Check whether Android WiFi is turned on 4. Check whether the Android WiFi is connected correctly 5. Check whether it is in the initialization state and wait for a while 6. Check whether the signal cannot be received due to too far distance or metal shielding
2	Connecting to Server, please check the WiFi signal!	1. Check whether the robot is in the initialization state and wait for a while 2. Check whether the emergency stop button on the control cabinet is pressed 3. Check whether the emergency stop on the APP is pressed 4. Check whether the removable emergency stop box originally placed inside the control cabinet is pressed 5. Check whether the network cable from the robot controller in the control cabinet to the server circuit board is connected normally
3	Servo driver error	Restart the bot
4	Exceed software	Open and drag, and drag to the normal working area in

	limit position	reverse
5	Collision error	Quickly clear the error, and after a long time, it will become the error in point 3 above
6	Singular position	Change the torch orientation and re-teach
7	Slave station error	1. Restart the robot 2. Check whether the robot heavy-load line is connected correctly
8	Command error	1. Check whether the end point of the arc point is entered correctly 2. Check whether the current tool coordinate system of the robot is tool1, and whether the tool coordinate system is correctly entered
9	IO error; Arc start or end error	1. Check and re-clean the surface of the arc point 2. Check the workpiece grounding 3. Check whether the welding machine is turned on 4. Check the remaining welding wire in the wire feeder 5. Check whether the welding machine is alarmed and faulty
10	File error	Restart the bot
11	Params error	Restart the bot
12	Joint configuration warning	Restart the bot
13	Emergency stop!!!	1. Check whether the emergency stop button on the control cabinet is pressed 2. Check whether the emergency stop on the APP is pressed 3. Check whether the removable emergency stop box originally placed inside the control cabinet is pressed

9.2 MEGMEET Ehave2 CM500R Welding machine failure list and solutions

Quoted from the Megmeet Ehave2 CM500R User Manual, with a focus on the following.

Welding power fault code and the reasons and solutions are showed in the table

5-1 below.

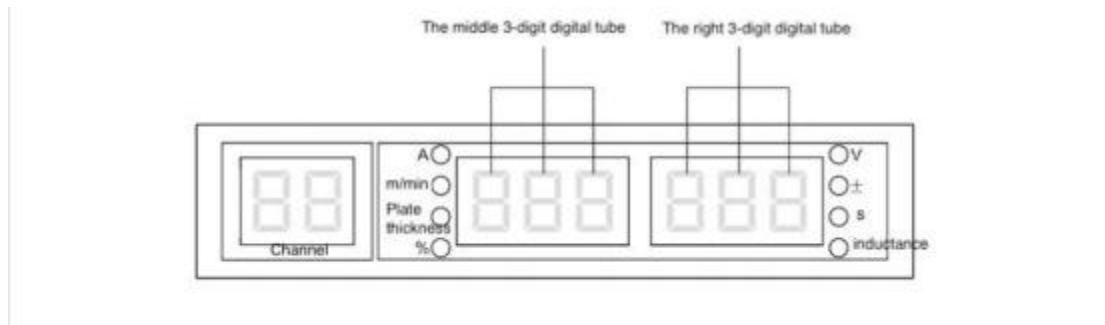


Figure 5-1 Fault Codes Display

Table 5-1: Welding Power Supply Fault Codes and Solutions

Fault Code	Display Content		Fault Description	Possible Causes	Solutions
	The left 3 tube	The right 3 tube			
E1	E1	None	Welding torch fault	Welding torch switch closed when power is turned on Welding torch switch damaged;	Set welding torch switch to OFF; Replace welding torch switch
E2-1	E2	1	Fan Fault	Fan power short circuit	Check if fan is damaged; Check fan drive line for shorts
E3-1	E3	1	Input power abnormal	Input undervoltage	Check input cable connection; Verify input power
E3-2	E3	2	Input power abnormal	Input overvoltage	Check input cable connection; Verify input power
E3-3	E3	3	Input phase loss	Excessive input bus ripple	Check input cable connection; Verify input power
E4-1	E4	1	Power supply heat abnormal	Primary IGBT module overtemperature	Operate within duty cycle rating; Check ventilation; Clean heatsink; Verify fan operation; Check IGBT module
E4-2	E4	2	Power supply temperature abnormal	Secondary diode module overheat	Operate within duty cycle rating; Check ventilation; Clean heatsink; Verify fan operation; Check diode module

E5	E5	1~17	Key error	Key stuck during assembly; Key jammed	Refer to Figure 3-1 to check corresponding key (see right 3-digit display)
E6	E6	None	Output overcurrent	Output short circuit or excessive current	Check for output shorts
E7-1	E7	1	Communication fault	Display-control board communication error	Check communication cable between display and control boards
E7-2	E7	2	Communication fault	Control board-wire feeder display communication error	Check communication cable between control board and wire feeder display
E7-3	E7	3	Communication fault	Control board-robot communication box error	Check communication cable between control board and robot communication box
E8	E8	None	Output overvoltage	Input voltage too high; Main transformer damaged; Wrong output wiring	Check input voltage; Inspect main transformer; Verify output wiring
E9	E9	None	Primary Overcurrent	IGBT failure; Main transformer damage; Diode module failure	Check power board; Inspect main transformer; Test diode module
E10	E10	None	Primary Overvoltage	Insufficient input power; Input voltage too high	Verify input power capacity; Check input voltage
E11	E11	None	Current Hall Sensor Disconnected	Current Hall sensor disconnected	Reconnect Hall sensor
E12-1	E12	1	Software version mismatch	Control board version mismatch	Update firmware or contact manufacturer
E12-2	E12	2	Software version mismatch	Wire feeder display version mismatch	Update firmware or contact manufacturer
E12-3	E12	3	Software version mismatch	Robot communication box version mismatch	Update firmware or contact manufacturer
E12-4	E12	4	Software version	Group control	Update firmware or contact manufacturer

			mismatch	communication box mismatch	
E13-1	E13	1	Motor drive fault	Average motor current overlimit	Check if wire feeding is blocked or stuck; Clean/replace liner; Clean/replace contact tip
E13-2	E13	2	Motor drive fault	Peak motor current overlimit	Check motor power line shorts; Inspect 7-core control cable
E14-1	E14	1	Solenoid short circuit	Solenoid overcurrent	Check solenoid for shorts/damage
E14-2	E14	2	Solenoid open circuit	Solenoid open/disconnected	Check solenoid continuity and wiring
E15-1	E15	1	Auxiliary power fault	High-voltage auxiliary power failure	Contact service
E15-2	E15	2	Auxiliary power fault	Control board auxiliary power failure	Contact service
E16	E16	None	Communication box failure	Robot interface board recognition failed	Contact service
E18-1	E18	1	E-stop fault	Special machine e-stop signal error	Check communication cable; Inspect special machine
E18-2	E18	2	E-stop fault	Robot e-stop signal error	Check robot communication cable; Inspect robot
E19-1	E19	1	Robot communication fault	Robot-welder communication error	Check digital comm ports and cables
E20-1	E20	1	Unregistered board	Display board unauthorized	Purchase Original manufacturer authorized board
E20-2	E20	2	Unregistered board	Control board unauthorized	Purchase Original manufacturer authorized board
E21	E21	None	Heat value fault	Heater power out of range	Verify heater valve specifications Check heater valve wiring for shorts
E22	---	---	No craft matched	No matching craft	Select craft which is in the craft table
E26-1	E26	1	Water tank fault	No flow detected in water tank	Check whether the water flow loop is smooth; Check whether the water pipe is bent;

					Check whether the water tank is operating normally;
E27-1	E27	1	JOB number out of range	Robot JOB number exceeds 1-99	Check whether the job issued by the robot teaching pendant complies with the specification 1-99
E27-2	E27	2	JOB number out of range	Group control JOB number exceeds 1-99	Check whether the job issued by the group control system complies with the specification 1-99
E29	E29	None	Group communication fault	Welder-cluster controller communication error	Contact service
E30	E30	None	Group parameter out of range	The craft parameters cluster control sent to welder out of range	The cluster control system sends the correct craft parameters

10. SmartPlus Components

10.1 Intelligent Dragger

Intelligent Dragger (V1.0.0) has 5 buttons, three on the front and two on the back. Functionally, the three on the front are "Safe point", "Line Point" and "Arc Point" from top to bottom, and the two on the back are "Start" and "Drag".

1. When the "Safe point" button is pressed, the movement to the current position will be recorded in the form of safe point movement.

2. When the "Line Point" button is pressed, the movement to the current position will be recorded in the way of line welding.

3. When the "Arc Point" button is pressed, it will record the movement to the current position in the way of arc welding, note that when pressing the first time, it is the middle position of the arc, and when the second time is pressed, it is the end position of the arc, and so on.

4. When the "Start" button is pressed, welding will be started immediately, and the robot should be dragged to the vicinity of the position recorded in the first command before pressing the button.

5. When the "Drag" button is pressed, it will immediately switch between drag and non-drag modes.

6. When any button is pressed during the welding craft, the welding will stop

immediately.

7. Different from using software to start welding, the welding initiated by the intelligent dragger will be cleared (default) regardless of any problems encountered or successfully completed and can be modified on the settings page after the software version V1.2.3.